Vishank Singh

ai23btech11032@iith.ac.in | vishanksinghh@gmail.com | +91 8178882876 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.51
Seth Anandram Jaipuria School	Class XII (CBSE)	2023	95.80%
Seth Anandram Jaipuria School	Class X (CBSE)	2021	96.20%

SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, SQL, CUDA

Libraries & Frameworks: Numpy, Matplotlib, PyTorch, cvxpy, HuggingFace transformers, ReactJS

Technologies: Git, GitHub, Linux, Docker

RELEVANT COURSES

AI: Introduction to Artificial Intelligence, Programming for AI, Foundations of Machine Learning, Deep Learning, Reinforcement Learning, Online Learning, Robotics, Computer Vision, Natural Language Processing, Multi-Agent Systems

Math: Differential Equations, Series of Functions, Probability and Random Variables, Introduction to Metric Spaces, Applied

Statistics, Convex Optimization, Numerical Analysis

Computer Science: Programming, Discrete Math, Data Structures, Computer Architecture, OS-1, DBMS-1, Algorithms, Compilers-1, DBMS-2, Software Engineering, Data Mining, GPU Programming, Distributed AI Training

Electrical Engineering: Vector Calculus, Matrix Theory, Linear Systems and Signal Processing, Control Systems

EXPERIENCE

Summer Analyst, Goldman Sachs (Data Engineering)

- Developed an agent to streamline the update and maintenance of an internal financial product JSON database.
- Created a tool loop and a deterministic update engine to minimize non-determinism propagation.
- Successfully pushed the feature to production, enhancing operational efficiency.

Teaching Assistant, CS 2323 Computer Architecture

- Maintained the riscv rv64imfd zicsr simulator and the associated Visual Studio Code extension for teaching purposes.
- Developed both the simulator and the VSCode extension to enhance the learning experience of RISC-V assembly language.
- Collaborated with faculty to ensure effective integration of tools into the curriculum, fostering a deeper understanding of assembly language concepts.

PROJECTS

CRASM (CUDA RISC-V Assembler) (GitHub)

- Engineered CRASM, a high-throughput RISC-V assembler utilizing modern CUDA C++ to execute assembly pipelines entirely on NVIDIA GPUs.
- Architected parallel compute acceleration strategies to translate instructions concurrently, maximizing GPU utilization and overall assembly throughput.

- Benchmarked system performance across multiple file sizes, achieving a 10-26x speedup on million-scale workloads (1-50M lines) compared to standard CPU-based assemblers (llvm-mc, GNU).

MLIR-Based Compile-Time Neural Network Compiler (GitHub)

- Designed and built a custom C++ compiler driver using Clang AST Matchers to intercept neural network written in C++ and lower them into a custom MLIR dialect representation.
- Developed a custom LLVM/MLIR optimization pass that performs static evaluation and JIT interpretation of SGD training loops at compile-time, rewriting functions to bake trained weights directly into static constants in the final binary.

Extension of BANG for Dynamic Graph Approximate Nearest Neighbor Search (GitHub)

- Extended the BANG (Billion-scale Approximate Nearest Neighbour Search) framework to support dynamic graphs by enabling efficient incremental updates (edge/vertex insertions and deletions) while preserving high-recall ANN performance on a single GPU.
- Designed and implemented mechanisms for maintaining a compressed graph index and distance computations under evolving graph topology, avoiding full rebuilds and reducing update overhead compared to static recomputation.
- Evaluated the dynamic extension on large-scale datasets, demonstrating sustained throughput and recall under continuous graph modifications, building on BANG's host-GPU hybrid architecture for billion-scale ANNS.

RISC-V Virtual Machine (GitHub)

- Implemented in C++ a VM supporting RV64IMFD, zicsr extension, syscalls.
- Supports efficient debugging through command line and generates machine code.
- Memory mapped IO devices (Display) are supported and can be extended easily.

Visual Studio Code Debugger Extension for RISCV (GitHub)

- Implemented a VS Code extension in Typescript for language and debug support for RISC-V assembly in VS Code.
- Implemented a Debug Adapter supporting run, undo, redo, breakpoints, memory view, register view/edit.
- Supports syntax highlighting, code completion, linting, disassembly natively.

Tensor computation library (GitHub)

- Implemented a tensor computation library in C++ with CUDA support for GPU.
- Supports basic tensor arithmetic, broadcasting and memory management.
- Basic automatic differentiation support via computational graphs.